

미분기하학과 일반상대론(이) (PHY309-01)

학기	2026년 2학기
담당교수	Carlos Scarinci
이수구분	교선평수
학점	3.0
강의시간	화09:00-10:30(E7 - L21), 목09:00-10:30(E7 - L21)
강의실	
소속	기초학부

교과개요

This course will introduce Einstein's theory of gravitation from a mathematical point of view. Basic concepts of differential geometry will be developed alongside basic concepts of general relativity. The main focus of the course will be on the definition of mathematical concepts, the study of their properties and relations, and their illustration in terms of physical examples.

교과목표

After completing this course, the students should have a clear understanding of fundamental concepts of differential geometry (manifolds, metrics, connections, geodesics and the curvature tensor) and general relativity (spacetime and causality, the equivalence principle and the Einstein field equations), their relation, and their application in the description of basic models in cosmology.

학습윤리

Cheating and plagiarism are serious offenses and will not be tolerated. Any student caught plagiarizing or cheating will automatically fail the course. This also applies to students who knowingly allow their work to be copied.

교과정책(방침)

Attendance of the midterm and final exams are mandatory (absence in either exam will result in a failing grade). Make-up exams might be granted upon request, in a case-by-case basis and with appropriate documentation explaining each circumstance.

준비물 및 기타

Prerequisites: Multivariable Calculus and Linear Algebra